# DSA Array Reference: Java vs JavaScript

A syntax and function guide for competitive programming and technical interviews.

## 1. Initialization & Basics

| Operation | Java (Static Arrays & ArrayList) | JavaScript (Dynamic Arrays) |
|---|---|---|
| **Declare Empty / Fixed** | `int[] arr = new int[5];`<br>`ArrayList<Integer> list = new ArrayList<>();` | `let arr = [];`<br>`let arr = new Array(5).fill(0);` |
| **Initialize with values** | `int[] arr = {1, 2, 3};`<br>`List<Integer> list = Arrays.asList(1, 2, 3);` | `let arr = [1, 2, 3];` |
| **Get Length / Size** | `int len = arr.length;`<br>*(Property)*<br><br>`int size = list.size();`<br>*(Method)* | `let len = arr.length;`<br>*(Property)* |

## 2. Insertion & Deletion

| Operation | Java (ArrayList usually required) | JavaScript |
|---|---|---|
| **Add to End (Push)**<br>*O(1) amortized* | `list.add(val);` | `arr.push(val);` |
| **Remove from End (Pop)**<br>*O(1)* | `list.remove(list.size() - 1);` | `let last = arr.pop();` |
| **Add to Start (Unshift)**<br>*O(N) - shifts elements* | `list.add(0, val);` | `arr.unshift(val);` |
| **Remove from Start (Shift)**<br>*O(N) - shifts elements* | `list.remove(0);` | `let first = arr.shift();` |
| **Insert at Index `i`** | `list.add(i, val);` | `arr.splice(i, 0, val);` |
| **Remove at Index `i`** | `list.remove(i);` | `arr.splice(i, 1);` |

## 3. Searching & Accessing

| Operation | Java | JavaScript |
|---|---|---|
| **Access Element at `i`** | `int val = arr[i];`<br>`int val = list.get(i);` | `let val = arr[i];` |
| **Update Element at `i`** | `arr[i] = val;`<br>`list.set(i, val);` | `arr[i] = val;` |
| **Find Index of Value**<br>*O(N) linear search* | `list.indexOf(val);`<br>*(Returns -1 if not found)* | `arr.indexOf(val);`<br>*(Returns -1 if not found)* |
| **Check if contains** | `boolean exists = list.contains(val);` | `let exists = arr.includes(val);` |

## 4. Sorting & Reversing

| Operation | Java | JavaScript |
|---|---|---|
| **Sort Ascending**<br>*O(N log N)* | `Arrays.sort(arr);`<br>`Collections.sort(list);` | `arr.sort((a, b) => a - b);`<br>*JS default sort is alphabetical! Always pass the comparator for numbers.* |
| **Sort Descending** | `Arrays.sort(arr, Collections.reverseOrder());`<br>*(Objects only, not int[])*<br><br>`Collections.sort(list, Collections.reverseOrder());` | `arr.sort((a, b) => b - a);` |
| **Reverse Array** | *Manual loop for primitives:*<br><br>`for(int i=0; i<n/2; i++) { swap(arr, i, n-1-i); }`<br>`Collections.reverse(list);` | `arr.reverse();`<br>*(Mutates original array)* |

## 5. Subarrays & Copying

| Operation | Java | JavaScript |
|---|---|---|
| **Clone / Copy Entire Array** | `int[] copy = arr.clone();`<br>`List<Integer> copy = new ArrayList<>(list);` | `let copy = [...arr];`<br>*(Spread operator)*<br><br>`let copy = arr.slice();` |
| **Get Subarray (from `i` to `j-1`)** | `int[] sub = Arrays.copyOfRange(arr, i, j);`<br>`List<Integer> sub = list.subList(i, j);` | `let sub = arr.slice(i, j);` |